TEXORA · LECTURE SHEET

Long Staple: Manufacture of Jute Blended Yarns

Jute is a long bast fibre that cannot be handled on cotton machinery. This sheet follows it from bale selection and emulsion softening through breaker and finisher cards, gill drawing and the jute spinning frame, and then shows how it is blended with other fibres and counted in its own lb/spindle system.

YM-4201 · MODERN YARN MANUFACTURING · TOPIC 4 · LEVEL 4 TERM 2 · YARN MANUFACTURING SPECIALIZATION

1. Jute as a long, bast fibre

Jute is obtained from the stem bark of *Corchorus* plants, principally *C. capsularis* (white jute) and *C. olitorius* (tossa jute). It is a **bast fibre**: the useful fibres lie in the phloem layer outside the woody core and are recovered by retting — controlled microbial rotting of the stem in water — followed by stripping, washing and drying. What reaches the mill is not a single-cell fibre like cotton. The commercial “fibre” is a strand, typically 1.5–3.5 m long, made of many short ultimate cells, each only about 2–5 mm long and roughly 15–25 µm across, cemented together by lignin and hemicellulose. That structure explains almost every feature of jute processing.

- **Long staple, in strands.** Because the strand is metres long, jute belongs to the long-staple family of spinning systems, alongside flax and worsted wool. It must be opened, carded and drafted by pinned rollers and fallers that can control long material, not by cotton saw-tooth clothing and short roller settings.
- **Stiff and lignified.** Jute contains appreciably more lignin than cotton. It is strong but comparatively inextensible, harsh when dry, and it breaks rather than drafts if it is worked without softening. Moisture and a small amount of oil transform its handling.
- **Uneven by nature.** Strand fineness, colour and strength vary with plant variety, retting and the position in the stem — root ends are coarser and harder than middle and tip portions. Careful batching, in which grades are mixed deliberately, is therefore the first quality operation, not an administrative detail.

PROPERTY	JUTE (TYPICAL BEHAVIOUR)	CONSEQUENCE IN SPINNING

Form delivered	Long strands of short ultimate cells bonded by lignin	Needs softening and progressive splitting on cards
Length	Strands commonly 1.5–3.5 m; ultimate cells 2–5 mm	Long-staple machinery, pinned drafting control
Handle dry	Stiff, harsh, low extension	Emulsion softening and piling before carding
Strength	Good tenacity, falls when over-retted or degraded	Grade and batch selection matter
Blend partners	Cotton, viscose, flax, wool, synthetics	Softer, finer or cheaper yarns for wider end uses

2. Batching and batch selection

A **batch** is the quantity of jute of a chosen mixture of grades prepared for a particular yarn quality and count. Batch selection is done bale by bale and bundle by bundle, judging colour, strength, fineness, root content and defects. The principles are:

1. **Match the batch to the yarn.** Finer, stronger yarns demand longer, finer, better-retted fibre with little hard root. Coarse sacking yarns can accept lower grades. Using fine fibre for a coarse yarn wastes money; using poor fibre for a fine yarn creates ends down and uneven yarn that no later setting can repair.
2. **Mix for evenness.** No single bale is uniform. A batch deliberately combines bales and grades so that variations average out over the whole production run. Sudden changes of batch in the middle of an order show up as count and shade variation.
3. **Cut off hard ends.** The woody root end (the “cuttings” end) is cut away where the specification requires it; cuttings are not discarded from the mill — they are directed to coarser products.
4. **Record the recipe.** The proportions of each grade in the batch are recorded, so a successful quality can be repeated and a faulty one traced.

3. Softening with emulsion

3.1 Why softening is needed

Dry jute strands are stiff bundles. On the card the pins must split the bundles into finer filaments and separate entangled material; dry, lignified bundles resist splitting, break into short pieces and give a harsh, hairy, uneven sliver. Softening plasticises the strand with water and lubricates it with a thin film of oil, so it bends, splits and slides instead of breaking.

3.2 The emulsion

The softening agent is an **oil-in-water emulsion**: fine droplets of a batching oil dispersed in water, stabilised by an emulsifier, applied on a softener machine — a pair or series of fluted rollers that squeeze the emulsion into the strands as they pass through — or, in simpler layouts, by spraying during conveying. The water provides the moisture that softens; the oil remains after the water is absorbed or dries, lubricating fibre against fibre and fibre against metal.

Typical values must be read as typical mill values, not as a universal recipe, because oil type, fibre grade and end use all shift them. As a guide, emulsion is commonly applied at roughly 25–35% on the weight of fibre, the emulsion itself contains roughly 15–20% oil, and the result is roughly 4–6% oil on the weight of fibre. Excess oil is as faulty as too little: it loads the card clothing, causes roller lapping and can stain and smell in the finished goods; too little leaves the fibre harsh and increases breakage and waste.

> **WORKED EXAMPLE — EMULSION AND OIL ON FIBRE**
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> A batch of 1,000 kg of jute receives emulsion at 30% on fibre weight, and the emulsion contains 18% oil.
>
> Emulsion applied = 1,000 × 30/100 = **300 kg**.
>
> Oil applied = 300 × 18/100 = **54 kg**, which is 54 / 1,000 × 100 = **5.4% oil on the weight of fibre**.
>
> Water (and emulsifier) in the emulsion = 300 − 54 = **246 kg**.
>
> The figures sit inside the typical ranges stated above, so the recipe is plausible; a result of, say, 15% oil on fibre would signal an error of recipe or of calculation.

3.3 Piling and conditioning

After application the jute is **piled** and left, typically for 24–48 hours. During piling the emulsion penetrates the strand and moisture equalises, so the fibre reaches the card uniformly soft. Shortening piling is a false economy: unconditioned jute breaks more in carding and gives uneven sliver. Card promptly, in batch order, after piling.

4. Carding: breaker card and finisher card

Jute carding is normally done in two stages, because the change from stiff retted strand to a controlled sliver is too large for one machine.

4.1 Breaker card

Softened strands are fed to the **breaker card**, whose large pinned rollers (main cylinder with workers and strippers) tear the strands apart, begin splitting the bundles, remove loose dirt and deliver a first,

still coarse sliver. The action is deliberately opening rather than fine carding: forcing a finer action here would break fibre.

4.2 Finisher card

Several breaker slivers are combined as feed to the **finisher card**, whose finer pinning and closer settings split bundles further and improve parallelisation. Doubling at the finisher is itself an evening step: thick places in one sliver are diluted by the others. The finisher delivers the sliver for drawing. Opening must always be progressive, coarse to fine.

5. Drawing frames: push-bar and gill drafting

Card sliver is too uneven and its fibres too poorly parallel for spinning. Drawing in jute is done on frames whose drafting field is controlled by pinned fallers — the **gill** principle — because the fibres are long. Two families are met:

- **Push-bar drawing frame.** Faller bars carrying pins are pushed along a track through the drafting zone and returned underneath. The pins penetrate the sliver, carry the fibres forward at feed speed and control floating fibres until they are gripped by the delivery rollers.
- **Screw-gill (gill box) forms.** Fallers are driven by screws, giving a positive, regular faller movement widely used in long-staple drawing. Rotary and chain gills apply the same pinned-control idea with different bar mechanisms.

A jute line normally uses successive drawing passages — commonly described as first, second and third drawing — each with doublings and a modest draft. Draft thins the sliver; doubling averages its irregularities. Draft at any passage follows the same mass rule used throughout spinning:

$$\text{Output sliver weight} = \text{feed sliver weight} \times \text{doublings} / \text{draft}$$

> **WORKED EXAMPLE — A DRAWING PASSAGE**
>
> A drawing passage is fed with sliver of 85 g/m, uses 6 doublings and a draft of 7.5.
>
> Output weight = 85 × 6 / 7.5 = 510 / 7.5 = **68 g/m**.
>
> Note what the numbers mean: although six slivers entered, the output is lighter than one feed sliver, because the draft (7.5) is larger than the doublings (6). If draft had equalled doublings, the output weight would have stayed at 85 g/m while its evenness improved.

6. The jute spinning frame and winding

On the jute spinning frame the final drawn sliver — classical jute spinning often has no separate roving stage — is drafted under long-fibre control, attenuated to yarn size and given twist by a flyer as it winds

onto a bobbin. Twist is set for adequate strength and a firm package without harshness. Yarn then proceeds to winding, twisting for plied yarns or twine, reeling or spooling, and packing, according to end use.

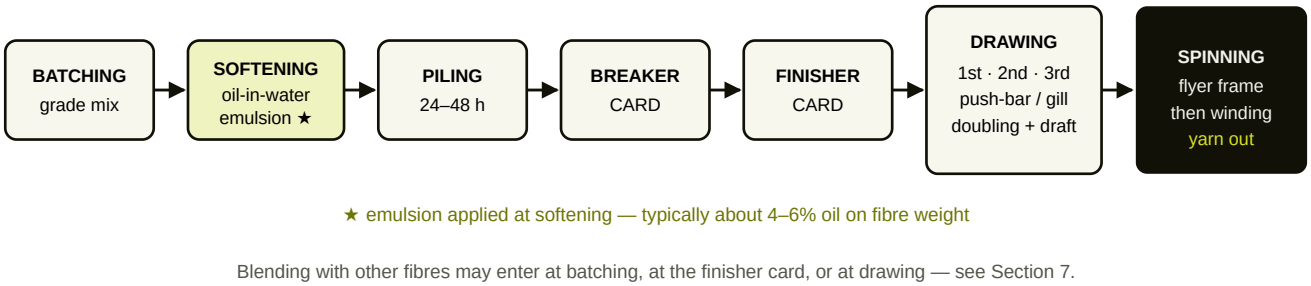


Figure 1 — Jute processing line. Batching, emulsion softening and piling prepare the long bast strands for progressive carding, pinned drawing and flyer spinning.

7. Blending jute with other fibres

Pure jute serves its traditional markets well, but blending widens the range in both directions: a softer or finer partner improves handle and allows finer counts, while jute itself brings bulk, strength and low cost to a blend. Common partners and their purpose are:

PARTNER FIBRE	WHAT IT CONTRIBUTES	TYPICAL REASON FOR THE BLEND
Cotton	Softness, finer spinning, better comfort next to skin	Apparel and furnishing yarns with jute character at a finer count
Viscose	Softness, lustre, more even dyeing	Improved appearance and handle for decorative fabrics
Flax and other bast fibres	Compatible long-staple behaviour, natural appearance	Bast-on-bast blends for furnishing and technical uses
Wool	Resilience, warmth, bulk	Carpet and furnishing yarns
Synthetic fibres	Strength, abrasion resistance, controlled shrinkage	Technical yarns, twines and durable furnishing products

Where blending is done decides how intimate the mixture is. (i) *At batching*, whole grades or fibre lots are combined before softening — suitable for jute-with-jute blending and for coarse blends, but the partner must survive the emulsion and piling stages. (ii) *At the finisher card*, slivers or laps of the two components are fed together — a practical route when the partner has been prepared separately on its

own machinery. (iii) *At drawing*, component slivers are combined in counted proportions at a drawing passage — **sliver blending** — which gives the most controllable blend ratio, because the proportion is set simply by the number and weight of slivers of each component, and can be checked by weighing the output. In every route the governing rule is the same: the components must be compatible in length and processing behaviour, or the longer or slicker component will draft preferentially, the blend ratio in the yarn will drift from the ratio in the feed, and the fault will appear as streaks, strength variation and uneven dyeing.

Blend ratio is always by weight, and must be verified in the yarn, not merely in the feed.

8. The jute count system: lb per spindle

Jute yarn is counted in its own indirect system. The count is the weight in pounds of one **spindle** of yarn, where one spindle is a fixed length of **14,400 yards**. A higher lb/spindle number therefore means a *heavier, coarser* yarn — the number is a weight, so it rises with coarseness, unlike cotton count, where the number rises with fineness.

$$\text{Jute count} = \text{weight (lb) of 14,400 yd of yarn}$$

WORKED EXAMPLE — CONVERTING 8 LB/SPINDLE TO TEX

One spindle = 14,400 yd = 14,400 × 0.9144 = 13,167.36 m = 13.16736 km.

A count of 1 lb/spindle means 453.6 g of yarn in 13.16736 km.

Tex is grams per kilometre, so tex for 1 lb/spindle = 453.6 / 13.16736 = **34.4 tex** (34.45 tex before rounding).

Therefore 8 lb/spindle = 8 × 34.45 = **275.6 tex, about 276 tex**.

Check the direction: the jute yarn is coarse, so a tex value in the hundreds is expected. A conversion that returned, say, 4 tex would be wrong in direction as well as in size.

The conversion factor is worth memorising: **tex ≈ 34.45 × (lb/spindle)**, and conversely **lb/spindle ≈ tex / 34.45**. Thus a 200 tex yarn is 200 / 34.45 = 5.8 lb/spindle. Within the mill, length–weight work is done directly in the jute system: a package holding 2.5 lb of an 8 lb/spindle yarn contains 2.5 / 8 × 14,400 = 4,500 yd of yarn.

Glossary

Bast fibre — fibre obtained from the stem bark (phloem) of a plant, as with jute and flax. **Retting** — controlled rotting of stems in water to free the bast fibres. **Batch** — a selected mixture of grades prepared for one yarn quality. **Batching oil** — oil applied in emulsion to soften and lubricate jute before carding. **Piling** — resting softened jute so emulsion penetrates and moisture equalises. **Breaker**

card — first, coarse card that opens the strands. **Finisher card** — second card that refines splitting and delivers card sliver. **Gill / push-bar drawing** — drafting controlled by pinned faller bars for long fibres. **Spindle (jute)** — a fixed length of 14,400 yd used as the basis of the jute count. **Sliver blending** — combining component slivers in counted proportions at drawing.

Key points

• Jute is a long bast strand of short lignin-bonded cells — stiff when dry, so softening is essential, not optional.
• Batch selection matches grade mixture to yarn quality and averages out bale variation.
• Softening uses an oil-in-water emulsion; typical results are roughly 4–6% oil on fibre weight, followed by 24–48 h piling.
• Opening is progressive: breaker card first, finisher card second, with doubling at the finisher aiding evenness.
• Drawing uses pinned faller (push-bar / gill) control; output weight = feed weight × doublings / draft.
• Blending may occur at batching, at the finisher card or — most controllably — by sliver blending at drawing; ratios are by weight and must be verified in the yarn.
• Jute count is lb per 14,400 yd spindle; higher means coarser. tex ≈ 34.45 × lb/spindle.

Practice

1. Convert a jute yarn of 10 lb/spindle to tex. Then state, with a reason, whether it is coarser or finer than the 8 lb/spindle yarn worked in this sheet.
2. A mill wants 5.0% oil on the weight of fibre for a 500 kg batch. The emulsion contains 20% oil. Calculate the oil required and the total emulsion to be applied.
3. A package contains 2.5 lb of yarn of count 8 lb/spindle. Calculate the length of yarn in the package in yards, and then in metres.
4. Explain why jute is piled after emulsion application, and describe two faults that follow if piling time is cut severely.
5. Compare blending at batching with sliver blending at drawing: which gives the more controllable blend ratio, and why?

Answer key

1. Tex = 34.45 × 10 = **344.5 tex**. It is **coarser**: in the jute system the number is a weight per fixed length, so a larger number means more mass in the same length.
2. Oil required = 500 × 5.0/100 = **25 kg**. Emulsion = 25 / 0.20 = **125 kg** of emulsion to be applied.
3. Length = 2.5 / 8 × 14,400 = **4,500 yd**. In metres: 4,500 × 0.9144 = **4,114.8 m**.

4. Piling lets the emulsion penetrate the strand and lets moisture equalise, so the whole batch reaches the card uniformly soft and pliable. Cutting it severely gives (any two): increased fibre breakage on the breaker card; harsh, uneven sliver that carries through to uneven yarn; higher waste and ends down in spinning.
5. Sliver blending at drawing is more controllable. The ratio is set by the counted numbers and weights of component slivers entering together, and can be checked by weighing the combined output; at batching the components are mixed as loose fibre lots, so proportion and intimacy depend on manual mixing and on both components surviving softening and carding identically.
